

MERCED CASTLE, CA, USA

WMO: 724810

Lat: 37.383N

Lon: 120.567W

Elev: 58

StdP: 100.63

Time Zone: -8.00 (NAP)

Period: 90-19

WBAN: 23203

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.1	-0.1	-5.6	2.4	10.3	-2.9	3.0	9.1	12.0	12.4	10.4	11.7	0.5	140	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	39.2	20.9	37.8	20.2	36.8	19.7	22.3	35.6	21.5	34.6	20.7	33.7	4.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	13.1	23.6	17.4	12.5	22.5	16.3	11.7	22.8	65.2	34.9	62.4	34.9	59.7	33.9	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	7.9	6.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	7.8	10.8	13.4	16.5	20.7	23.7	26.8	25.7	24.0	18.8	12.4	7.9
	DBStd	7.24	2.39	2.79	2.91	3.49	3.58	3.32	2.59	2.74	2.55	3.51	2.96	2.80
	HDD10.0	189	74	21	7	1	0	0	0	0	0	0	9	78
	HDD18.3	1324	326	209	154	77	17	2	0	0	1	36	179	324
	CDD10.0	2893	7	45	112	195	331	410	522	487	419	272	81	13
	CDD18.3	986	0	0	1	21	90	162	263	229	170	49	1	0
	CDH23.3	13647	0	4	42	404	1320	2243	3601	3059	2219	718	37	0
CDH26.7	7639	0	0	2	144	639	1249	2225	1832	1254	288	6	0	
Wind	WSAvg	2.5	1.6	2.1	2.3	2.7	3.1	3.1	3.3	3.1	2.6	2.2	1.7	1.7
Precipitation	PrecAvg	314	64	58	48	23	14	3	0	0	1	20	26	56
	PrecMax	560	166	192	137	81	63	29	5	3	6	80	67	131
	PrecMin	108	0	6	0	0	0	0	0	0	0	1	0	3
	PrecStd	107	46	40	37	22	15	6	1	1	2	21	19	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	23.8	27.0	33.7	37.6	41.3	41.4	41.0	38.9	36.1	27.4	18.8
		MCWB	12.0	14.8	15.6	18.5	19.9	21.6	21.8	20.9	20.0	19.7	17.3	12.3
	2%	DB	16.6	21.2	24.1	30.5	35.0	37.9	39.4	38.7	37.4	32.5	23.8	17.0
		MCWB	12.1	14.3	15.0	16.9	18.3	20.7	21.3	20.2	19.2	18.5	15.1	12.2
	5%	DB	14.7	18.8	22.6	27.8	32.8	36.0	37.8	37.3	35.9	29.7	21.6	15.2
		MCWB	11.2	13.4	14.9	15.9	17.7	19.5	20.5	19.8	19.0	17.7	14.6	11.3
	10%	DB	12.9	17.2	20.8	25.3	30.5	33.3	36.3	35.4	33.5	27.4	19.0	13.7
		MCWB	10.3	12.5	14.4	15.3	17.0	18.7	20.2	19.3	18.5	16.4	13.4	11.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.8	17.3	18.0	21.9	21.2	22.8	24.0	22.5	21.2	21.9	18.6	16.3
		MCDB	15.9	21.1	21.6	27.7	35.3	37.1	38.7	37.6	35.5	30.6	22.9	16.5
	2%	WB	13.0	15.3	16.6	18.5	19.8	21.8	22.5	21.6	20.4	20.3	17.3	14.4
		MCDB	14.4	18.8	21.4	26.1	32.8	35.3	35.9	36.0	34.2	28.7	20.8	15.0
	5%	WB	12.1	14.2	15.8	17.3	18.5	20.7	21.6	20.7	19.8	18.9	16.2	13.0
		MCDB	13.7	17.6	20.9	24.8	30.4	32.5	34.9	34.1	33.0	27.3	19.8	13.9
	10%	WB	11.0	13.1	14.9	16.1	17.4	19.8	20.8	19.9	19.1	17.8	14.9	12.1
		MCDB	12.7	16.6	19.8	24.1	28.5	32.0	34.5	33.4	31.7	24.5	18.3	12.7
Mean Daily Temperature Range	5% DB	MDBR	9.8	11.9	12.8	15.5	17.0	17.8	18.8	18.7	19.0	16.4	14.1	10.8
		MCDBR	13.7	15.2	16.7	19.3	20.0	20.6	20.1	20.4	21.8	19.9	17.3	13.1
		MCWBR	8.7	9.1	8.4	7.8	7.1	6.9	7.0	6.9	6.8	7.8	8.9	8.5
	5% WB	MCDBR	8.9	12.7	14.1	17.4	18.3	18.9	18.4	18.8	19.5	17.2	13.9	8.6
		MCWBR	7.4	8.4	8.1	7.9	7.0	6.7	7.2	7.0	6.7	7.1	9.0	7.3
Clear-Sky Solar Irradiance	taub	0.347	0.349	0.356	0.358	0.362	0.352	0.366	0.352	0.354	0.362	0.351	0.330	
	taud	2.384	2.403	2.382	2.388	2.398	2.434	2.409	2.424	2.430	2.395	2.389	2.462	
	Ebn at Noon	828	879	907	921	919	924	908	916	897	851	815	826	
	Edh at Noon	90	100	112	117	118	114	116	112	105	99	89	79	
All-Sky Solar Radiation	RadAvg	2.33	3.42	4.90	6.29	7.53	8.38	8.16	7.40	6.08	4.41	2.92	2.13	
	RadStd	0.38	0.38	0.45	0.51	0.39	0.27	0.25	0.21	0.19	0.25	0.18	0.31	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (44 neighbors)	+0.40	N/A	-1.15	N/A	N/A	N/A	N/A	-114	+138	+35	

Nomenclature: See separate page